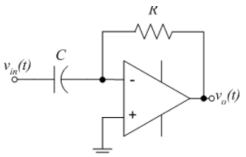
MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
Discipline Specific Elective (DSE) - MG4DSEPHY200 – SEMICONDUCTOR ELECTRONICS
ANSWER KEY

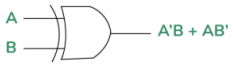
Part A

Answer any 10 questions. Each question carries 2 marks

1. Distinguish between JFET and BJT based on their input impedance and noise level?
   i. JFET: High input impedance, low noise
   ii. BJT: Low input impedance, higher noise
2. Define transfer characteristics of a DE-MOSFET?
   Relationship between drain current (I_D) and gate source voltage (V_{GS})
   Operation in both depletion and enhancement modes.
3. List the different types of ICs based on scale of integration.
   SSI (Small Scale Integration), MSI (Medium Scale Integration), LSI (Large Scale Integration), VLSI (Very Large Scale Integration)
4. What is meant by the term "summing amplifier" in op-amp applications?
   An op-amp circuit that adds multiple input voltages and gives a single output.
5. Why does an ideal op-amp draw no current at its input terminals, particularly at the summing junction?
   An ideal op-amp has infinite input impedance, so no current flows into terminals.
6. In which op-amp configuration the phase difference between input and output is out of phase?
   Inverting amplifier (180° phase difference)
7. Define an operational amplifier (op-amp). Why is it called so?
   An op-amp is a high-gain differential amplifier.
   It is used for mathematical operations like addition, subtraction, integration, etc.
8. Define an op-amp differentiator. Draw its basic circuit diagram.
   Circuit where output is proportional to rate of change of input



9. Which are the three basic gates? Give their symbols.
   Basic gates -AND,OR,NOT + Symbols
10. Give the truth table and symbol of XOR gate.



Truth Table

A (Input 1)	B (Input 2)	X = A'B + AB'
0	0	0
0	1	1
1	0	1
1	1	0

11. Write the 1's complement of 1101101_2

    Ans - 0010010

12. Write the output of a NOR gate when both inputs are 0.

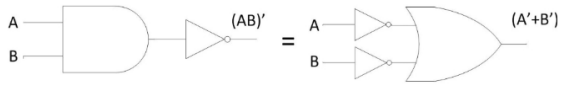
    Ans – 1

13. Simplify the Boolean expression (A+B)(A+C).

$$(A+B)(A+C) = A + AC + AB + BC \ldots.1 \text{ Mark}$$

$$= A\,(1 + B + C) + BC = \boldsymbol{A + BC} \ldots.1 \text{ Mark}$$

14. Illustrate De Morgan's first law using basic gates.



Part B

Answer any 6 questions. Each question carries 5 marks

15. Describe how reverse bias at the gate controls the drain current in JFET.

    i. No Gate Voltage ($V_{GS} = 0$),The channel width is maximum and maximum current flows. (0.5 Mark)

    ii. When reverse bias is applied, the depletion region widens and channel width reduces. Hence, drain current decreases. (1 Mark)

    iii. Increasing Reverse Bias Further, reduces the channel width and drain current becomes very small. (1 Mark)

    iv. At a certain negative gate voltage, the channel is completely closed and drain current becomes zero. (0.5 Mark)

    v. Schematic diagram – 2 Marks

16. Explain the effect of positive and negative gate voltage on depletion MOSFET operation.

    D-MOSFET can operate in both depletion and enhancement modes, depending on whether the gate voltage (V_{GS}) is negative or positive. (1 Mark)
    Depletion Mode-Applying negative voltage to the gate, Repels electrons from the channel. Channel becomes narrower. Drain current decreases. (1 Mark)
    Enhancement Mode-Applying positive voltage to the gate. Channel becomes wider and more conductive. Drain current increases. (1 Mark)
    Schematic Diagrams- 2 Marks

17. Explain the construction and working of n-channel enhancement only MOSFET
    Schematic diagram: p-substrate, n^+ source & drain, insulated gate (SiO_2), no channel. -3 Marks
    Working: At $V_{GS} = 0$ → OFF; when $V_{GS} > V_T$ → channel forms → current flows. 2 marks

18. Explain 2's complement subtraction using suitable examples.
    Explanation of 2s complement method (To subtract *B* from *A* using 2's complement:Find 2's complement of B, Add it to A ,If there is a carry, discard it → result is positive, If no carry, take 2's complement of result and add minus sign → result is negative) – 3 Marks
    Any ONE example -2 marks

OR Any TWO examples of different types 5 Marks

19. Give a brief explanation of digital integrated circuits, highlighting their practical applications.

Digital ICs are semiconductor devices that contain logic circuits on a single chip. -1.5Mark
Advantages -Small size, reliable Reduced cost, Low power, high speed -1.5 Mark
Applications -Computers, Communication, Consumer electronics, Automation, Medical (2 Marks)

20. Add 34 and 89 after converting to binary equivalent. Check your answer in decimal system.

Convert to Binary- 2 Marks

$$34 = 100010_2$$
$$89 = 1011001_2$$

Add the Binary digits – gives 1111011_2 - 2 Mark
Convert Result to Decimal-1111011_2=64+32+16+8+2+1=123- 1 Mark

21. What are monolithic ICs? Explain the fabrication of transistors and capacitors in a monolithic IC.

Monolithic ICs are integrated circuits in which all components (transistors, resistors, capacitors, etc.) are fabricated on a single silicon chip. (1 Mark)
Transistor Fabrication
Diffusion in n-type epitaxial layer on p-substrate, SiO_2 etched for selective regions
Forms emitter, base and collector, metal contacts added (2 Marks)
Capacitor Fabrication
Formed using reverse biased P–N junction, P and N regions → plates, depletion region → dielectric (2 Marks)

22. Construct the truth table for the Boolean expression $F = (A + B)(\bar{A} + C)$.Also explain how the output changes for different combinations of input variables.

A	B	C	A+B	Ā	Ā + C	F = (A+B)(Ā+C)
0	0	0	0	1	1	0
0	0	1	0	1	1	0
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	1	0	0	0
1	0	1	1	0	1	1
1	1	0	1	0	0	0
1	1	1	1	0	1	1

… 4 Marks

Output depends on two conditions:

1. $A + B = 1$→ at least one of A or B is 1
2. $\bar{A} + C = 1$→ either A = 0 or C = 1

F = 1 only when both conditions are satisfied- 1 Mark